

Policy Gradient Fine-Tuning of Diffusion Models with Timestep-Aware Optimization

Anonymous ECCV 2026 Submission

Paper ID #14843

Abstract. Despite their remarkable generative performance, diffusion probabilistic models are typically optimized via likelihood-based surrogates rather than the objectives dictated by their ultimate downstream goals. In this context, policy-gradient fine-tuning enables adaptation to arbitrary, non-differentiable reward functions. However, its standard formulation uniformly aggregates gradients across all denoising time-steps at every update, resulting in substantial computational and convergence inefficiencies. In this work, we investigate how individual time-steps contribute to the policy-gradient objective and show that this full-trajectory evaluation may be improved. Aiming at that, we introduce a generalized policy-gradient estimator defined over a sampled subset of timesteps and propose timestep-aware sampling strategies that prioritize reward-relevant regions of the denoising trajectory. We define two reward-dependent importance measures, reward sensitivity and reward variance, that quantify how the corruption at each timestep affects downstream rewards. These metrics reveal strongly non-uniform and reward-dependent timestep contributions. Across empirical comparisons, we show that informed timestep allocation improves efficiency. In particular, combining structural Signal-to-Noise Ratio (SNR) weighting with reward-aware sampling yields over $3\times$ faster convergence using only 40% of the timesteps on the CelebA-HQ benchmark. It demonstrates that policy-gradient fine-tuning benefits from accounting for the non-uniform and reward-dependent structure of the diffusion trajectory.

Keywords: Policy gradient fine-tuning · Timestep-aware optimization · Diffusion probabilistic models

1 Introduction

Diffusion probabilistic models [18, 37] have become a dominant framework for diverse generative modeling tasks in the past few years [8, 15, 19, 27, 30, 34, 50]. Despite their success, these models are typically trained with maximum likelihood on massive uncensored datasets, ensuring fidelity to the data distribution but overlooking task-specific objectives such as aesthetic quality, alignment, or fairness [32, 49].

To adapt diffusion models to such downstream goals, several fine-tuning strategies have been proposed, including supervised fine-tuning on curated datasets [7, 11, 33, 35] and inference-time guidance [2, 4, 13, 14, 29]. Policy gradient

fine-tuning [1, 12, 32, 49] is another relevant approach that frames the denoising trajectory as a multi-step decision-making process and enables direct optimization of non-differentiable black-box reward functions. It builds on prior reward-based methods, achieving strong results in areas such as human preference modeling [12], prompt-image alignment [1], and fairness [32, 49].

Despite the notable empirical performance of policy-gradient methods for diffusion models, they are hampered by inefficiencies that remain a critical bottleneck. Specifically, we highlight two key deficiencies: **i) Computational inefficiency:** The policy-gradient objective requires evaluating and differentiating the model at every denoising timestep, making training expensive. While previous work mitigates computational costs by randomly subsampling timesteps, their strategy is empirical and lacks a principled analysis or justification [1, 49]. Consequently, there is no formal framework for prioritizing timesteps under a limited computational budget. **ii) Convergence inefficiency:** Because the reward is observed only after the full denoising process, existing methods assign the same scalar reward to all timesteps, implicitly treating them as equally important. In supervised diffusion training, timestep-dependent effects on learning dynamics are well-documented and shown as critical to efficient convergence [5, 17, 22, 41, 45]. However, timestep contributions have not been studied in policy-gradient fine-tuning, where the objective shifts from likelihood maximization to task-specific reward optimization. We argue that timestep importance in this setting is not only non-uniform but *reward-dependent*. Thus, disregarding this structure leads to optimization instabilities and poor sample efficiency.

Building on these observations, we investigate how different denoising timesteps contribute to the policy-gradient objective optimization and how this observed behavior may be employed to mitigate both computational and convergence inefficiencies. Therefore, our main contributions are:

- **Weighted Policy-Gradient Estimator:** We introduce a general policy-gradient formulation with explicit timestep weighting, providing a principled timestep-aware framework for fine-tuning. This formulation reduces computational overhead by replacing expensive full-trajectory evaluations with stochastic timestep sampling (Sec. 4.1).
- **Reward-aware Timestep Weighting:** We formalize two novel reward-dependent importance measures—reward sensitivity and reward variance—that quantify the role of individual timesteps in task-specific alignment, using them to prioritize sampling on high-impact regions of the denoising trajectory (Sec. 4.3).
- **Empirical Validation:** We provide a comprehensive comparative analysis of various sampling strategies, i.e., uniform, SNR-based [5], reward-aware, and hybrid approaches, against the full-trajectory baseline. Importantly, although SNR-based weighting was originally proposed for supervised diffusion training, we demonstrate its effectiveness in policy-gradient fine-tuning. Our results demonstrate that optimized timestep selection (SNR $\times$ reward-aware) achieves over $3\times$ faster convergence while utilizing only 40% of com-

putational resources, consistently outperforming prior uniform subsampling methods on the CelebA-HQ [20] benchmark (Sec. 5).

2 Related Works

We review the state-of-the-art in fine-tuning diffusion models using policy gradient approaches for reward-based optimization. Additionally, we examine the literature on improving supervised training by addressing the uneven contributions of timesteps in the diffusion process. These two groups of approaches are those closely related to our proposal, as, to our knowledge, no prior art has addressed policy gradient fine-tuning using a timestep-aware approach.

2.1 Fine-tuning diffusion models

Diffusion models have achieved remarkable generative performance across several tasks [8, 15, 19, 27, 30]. However, their standard maximum-likelihood training objective optimizes fidelity to large-scale data distributions rather than task-specific criteria such as aesthetics, alignment, or fairness. As a result, adapting diffusion models to downstream objectives has motivated a broad range of fine-tuning approaches, including supervised fine-tuning on curated datasets [7, 11, 33, 35], inference-time guidance via auxiliary inputs [14, 28], and representation-level control methods [2, 4, 13].

A complementary line of work defines objectives through reward functions that score generated samples, spanning prompt-encoder tuning [3], reward-based supervised fine-tuning [9, 24], reward-conditioned generation [43, 48], and differentiable [6, 44] or preference-based reward optimization [25, 40, 46]. While these methods can effectively steer generation, they are constrained by annotation cost, the need for differentiable or curated rewards, or increased inference-time overhead, motivating alternative approaches for optimizing arbitrary objectives.

Recent work has formulated diffusion fine-tuning as a reinforcement learning problem optimized via policy gradients [1]. In this framework, the denoising process is modeled as a multi-step Markov decision process [39], where each timestep is a decision contributing to the final image. A terminal reward is computed on that image, and REINFORCE-style [42] updates apply this signal across the denoising trajectory. DPOK [12] enhances this formulation by incorporating KL-regularization to limit updates relative to the base model while optimizing learned human preference rewards. At scale, Zhang et al. [49] demonstrate that policy-gradient fine-tuning can jointly optimize multiple objectives across millions of prompts, while Miao et al. [32] introduce batch-level diversity rewards based on statistical measures.

Despite their effectiveness, these methods apply the same terminal reward signal to every denoising timestep and aggregate log-probabilities uniformly across the trajectory, implicitly assuming that each step contributes equally to reward optimization. However, it remains unclear whether this assumption is valid, especially considering extensive evidence from supervised diffusion training that

shows timestep contributions are highly non-uniform. In contrast, our approach explicitly investigates timestep-dependent contributions to reward optimization and integrates them into a timestep-aware policy-gradient objective.

2.2 Non-uniform contribution of timesteps in supervised training

In supervised diffusion training, timestep-dependent learning dynamics are well documented and have motivated a range of timestep-aware optimization strategies. Among these, SNR-based loss reweighting has emerged as a particularly influential approach. Choi et al. [5] demonstrate that diffusion models capture different types of semantic content depending on the SNR. Specifically, they focus on coarse global structures at low SNR, perceptually significant details at intermediate SNR, and fine details at high SNR. This finding underscores the importance of paying more attention to mid- and low-SNR timesteps. Hang et al. [17] further demonstrate that disparities in denoising difficulty across SNR regimes induce conflicting gradients, and that reweighting losses inversely to a clamped SNR reduces this interference and accelerates convergence without degrading sample quality. Together, these works establish SNR as a principled signal for weighting timestep contributions in supervised diffusion training.

Beyond loss reweighting, other approaches address timestep imbalance through non-uniform sampling based on empirical measures of informativeness, learning progress, or distributional change across the diffusion process [21, 22, 41, 45, 51, 52]. Other timestep-aware architectures explicitly allocate capacity or specialization across timesteps to reduce gradient conflict and improve efficiency [26, 31, 47].

Despite these advances, existing studies focus on likelihood-based supervised diffusion training, where timestep importance is defined by each step’s direct contribution to data reconstruction. In policy-gradient fine-tuning, the objective is instead defined by a task-specific reward that is observed only at the terminal sample and uniformly propagated across the denoising trajectory. Consequently, timestep contributions are induced implicitly by the interaction between the reward signal and the denoising dynamics, and may differ substantially from their role in likelihood-based pretraining. To our knowledge, this work provides the first systematic analysis of timestep contributions under reward-based policy-gradient fine-tuning, along with a sound strategy to leverage these characteristics to improve optimization efficiency.

3 Preliminaries: Problem Statement

We consider a denoising diffusion probabilistic model (DDPM) [18], in which data samples are generated by reversing a Markovian noising process. In the forward process, a data point $x_0 \sim p_{\text{data}}$ is progressively corrupted by Gaussian noise according to a fixed variance schedule $\{\beta_t\}_{t=1}^T$, where t is a given timestep and T is the total number of timesteps. It produces a Markov chain $(x_0, \dots, x_T)$ that converges to an isotropic Gaussian. The generative model is defined by the

corresponding reverse Markov process, parameterized by a neural network ϵ_θ , which induces a tractable conditional distribution

$$p_\theta(x_{t-1} \mid x_t, c) = \mathcal{N}(\mu_\theta(x_t, t, c), \beta_t I),$$

where c is an optional conditioning variable (e.g., a text prompt), and μ_θ is computed from the noise prediction $\epsilon_\theta(x_t, t, c)$ using the standard DDPM parameterization.

Given a pretrained model and the noise schedule $\{\beta_t\}$, sampling proceeds by drawing $x_T \sim \mathcal{N}(0, I)$ and iteratively applying these reverse transitions until obtaining x_0 . This defines a stochastic denoising trajectory $\tau = (x_T, \dots, x_0)$ governed by differentiable transition densities $p_\theta(x_{t-1} \mid x_t, c)$, which induces a distribution over final samples x_0 . The goal of policy-gradient fine-tuning is to adjust θ so that this distribution maximizes a task-specific reward $r(x_0, c)$, which assigns a scalar score to the generated image x_0 and may be non-differentiable with respect to θ . This objective is formalized as

$$J(\theta) = \mathbb{E}_{c \sim p(c), x_0 \sim p_\theta(x_0 \mid c)}[r(x_0, c)]. \quad (1)$$

To enable this formulation within the reinforcement learning framework, the iterative denoising process is reframed as a multi-step Markov Decision Process (MDP) [1, 12, 16, 32, 49], with components defined as follows:

$$s_t \equiv (c, t, x_t), \quad a_t \equiv x_{t-1}, \quad \pi_\theta(a_t \mid s_t) \equiv p_\theta(x_{t-1} \mid x_t, c),$$

$$P(s_{t+1} \mid s_t, a_t) \equiv (\delta_c, \delta_{t-1}, \delta_{a_t}), \quad R(s_t, a_t) \equiv \begin{cases} r(x_0, c), & t = 0 \\ 0, & t > 0 \end{cases}, \quad P(s_0) \equiv (c, T, \mathcal{N}(0, I)),$$

where s_t and a_t represent, respectively, the state and action at timestep t , π_θ corresponds to the reverse diffusion policy, P defines the deterministic backward transitions from x_t to x_{t-1} , $P(s_0)$ initializes the process from pure noise $x_T \sim \mathcal{N}(0, I)$, and R assigns a terminal reward based on the final sample x_0 .

Within this MDP, the policy gradient for maximizing $J(\theta)$ follows the REINFORCE algorithm [39, 42]. By treating the log-probability of the trajectory as the object of differentiation, the gradient can be estimated using Monte Carlo samples, as per

$$\nabla_\theta J(\theta) = \mathbb{E}_{c \sim p(c), \tau \sim p_\theta(\tau \mid c)} \left[r(x_0, c) \sum_{t=1}^T \nabla_\theta \log p_\theta(x_{t-1} \mid x_t, c) \right], \quad (2)$$

where $\tau = (x_T, x_{T-1}, \dots, x_0)$ denotes the trajectory of noisy images produced by the reverse diffusion process under the policy $p_\theta(\cdot \mid \cdot, c)$.

4 Timestep-Aware Optimization Framework

4.1 Weighted Policy-Gradient Estimator

The policy-gradient in Eq. 2 requires summing over all T denoising timesteps at every update. To reduce computational cost, we replace the full summation

with stochastic sampling by drawing $t \sim p_\phi(t)$, where the sampling distribution is defined through a weighting function $w_\phi(t)$ as

$$p_\phi(t) = \frac{w_\phi(t)}{\sum_{t'=1}^T w_\phi(t')}. \quad (3)$$

This sampling scheme yields a generalized weighted estimator, analogous to Eq. 2, that expresses the policy gradient update as an expectation with respect to the timestep sampling distribution $p_\phi(t)$, as per

$$\nabla_\theta \tilde{J}(\theta) = \mathbb{E}_{c \sim p(c), \tau \sim p_\theta(\tau|c)} [\mathbb{E}_{t \sim p_\phi(t)} [r(x_0, c) \nabla_\theta \log p_\theta(x_{t-1} | x_t, c)]] , \quad (4)$$

where $\tilde{J}(\theta)$ denotes the timestep-weighted surrogate objective whose gradient is given by Eq. 4 (see Supp. Material). Under this weighted update, the choice of $w_\phi(t)$ allows computation to concentrate on specific timesteps according to a given criterion, thereby modulating their relative contribution to the policy-gradient update.

4.2 Reward-Agnostic Timestep Weighting

In diffusion models, the statistical properties of the denoising problem vary systematically across timesteps and are commonly characterized by the signal-to-noise ratio (SNR), which quantifies the relative strength of preserved data signal and injected noise. Following [23], the SNR at timestep t is defined as

$$\text{SNR}(t) = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}.$$

Prior work [5,17] has shown that SNR correlates strongly with both the optimization difficulty and the semantic content of different denoising stages. Such approaches have exploited these findings through SNR-based reweighting schemes that improve convergence and sample quality in supervised training.

We therefore adopt a *baseline SNR-based weighting* scheme, inspired by [5], as a reward-agnostic timestep importance weighting, given by

$$w_{\text{SNR}}(t) = \frac{1}{(k + \text{SNR}(t))^\gamma}, \quad (5)$$

where $w_{\text{SNR}}(t)$ is a concrete instantiation of the general weighting function $w_\phi(t)$ (Eq. 3), k stabilizes the weighting by preventing excessively large values at very small SNR, and γ controls the strength of down-weighting applied to high-SNR timesteps. This baseline importance function downweights high-SNR timesteps and reallocates computation toward noisier and semantically formative regions of the trajectory.

As expected, SNR reflects only the corruption level induced by the noise schedule, not the semantic relevance of each timestep for a specific task. Since different reward functions emphasize different visual attributes, the relative importance of timesteps for optimization can vary across objectives. This motivates going beyond reward-agnostic weighting and incorporating reward-dependent timestep structure, which we formalize in the next section.

4.3 Reward-Aware Timestep Weighting

Different reward functions emphasize different semantic attributes of the generated image, such as fine visual detail, object identity, or global structure. These attributes are consolidated at different stages of the denoising process, reflecting the fact that diffusion models encode different types of visual information at different noise levels. As a result, the relevance of a timestep for reward-optimization depends on the semantic properties emphasized by the reward function.

To quantify how individual timesteps affect task-specific rewards, we propose a procedure to measure how much reward-relevant information is determined at a given noise level. Given a set of N images $\{x_0^{(i)}\}_{i=1}^N$, we generate M independent corruptions at timestep t using the forward diffusion process,

$$x_t^{(i,j)} \sim q(x_t | x_0^{(i)}), \quad j = 1, \dots, M,$$

and denoise each corrupted sample deterministically with the pretrained model to obtain reconstructions

$$\hat{x}_0^{(i,j)} \sim p_\theta(\hat{x}_0 | x_t^{(i,j)}).$$

Each of the $N \times M$ reconstructions is then evaluated by the reward function, as illustrated in Fig. 1.

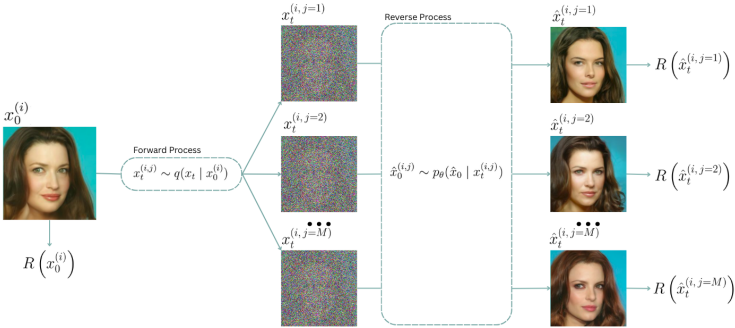


Fig. 1: Reconstruction-based procedure for measuring timestep contributions. For a timestep t and input image $x_0^{(i)}$, the forward diffusion process is applied M times to generate corrupted samples $x_t^{(i,j)}$. These are then denoised by the pretrained model to obtain reconstructions $\hat{x}_0^{(i,j)}$ that are evaluated by the reward function.

As the level of corruption increases, reconstructed images naturally deviate more from the originals, since progressively less information about the original content remains. This results in larger deviations and higher variability in the reconstructed samples. However, these visual degradations do not necessarily translate uniformly into changes in reward: some features lost at a particular

timestep may be irrelevant to the reward, while small changes in other features can cause large reward fluctuations (see Supp. Material).

However, our goal is to characterize timestep dependence from the perspective of a specific reward function rather than visual distortion alone. Using the data-generation procedure illustrated in Fig.1, we therefore introduce two metrics that quantify how much influence each noise level has on a given reward:

Reward sensitivity. This metric measures the average change in reward when an image is corrupted up to timestep t and then reconstructed:

$$\Delta R(t) = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{M} \sum_{j=1}^M |R(x_0^{(i)}) - R(\hat{x}_0^{(i,j)})| \right). \quad (6)$$

Reward variance. This metric measures the variability of the reward across reconstructions of the same image corrupted at timestep t :

$$\sigma^2 R(t) = \frac{1}{N} \sum_{i=1}^N \text{Var}(\{R(\hat{x}_0^{(i,j)})\}_{j=1}^M). \quad (7)$$

Together, these metrics capture how much a given timestep retains control over reward-critical attributes: low values indicate that the reward is already effectively determined at that noise level, so subsequent denoising steps have little leverage to change it, whereas high sensitivity or variance implies that, at that noise level, the model still has freedom to shape reward-relevant features and steer the reconstruction toward higher-reward outcomes. In our framework, these metrics yield timestep-indexed weighting functions, denoted $w_{\Delta R}(t)$ and $w_{\sigma^2 R}(t)$, effectively employed for timestep weighting through Eq. 3.

5 Experiments

Our experiments aim to provide empirical evidence that timestep sampling, when combined with structured non-uniform weighting, can improve the efficiency of policy-gradient fine-tuning. Specifically, we investigate whether it is possible to reduce the number of timestep evaluations per optimization step, thereby lowering computational cost, while simultaneously improving convergence speed by prioritizing timesteps that contribute more strongly to reward optimization.

To isolate this effect, we fine-tune a representative diffusion model under a fixed reward function and compare full-trajectory policy-gradient training with uniform sampling, SNR-based sampling, and our reward-aware strategies.

5.1 Reward-Aware Metrics Behavior

Before employing the *Reward sensitivity* and *Reward variance* metrics to guide timestep weighting in policy-gradient training, we empirically examine their behavior using a pretrained DDPM on the CelebA-HQ [20] dataset and a total

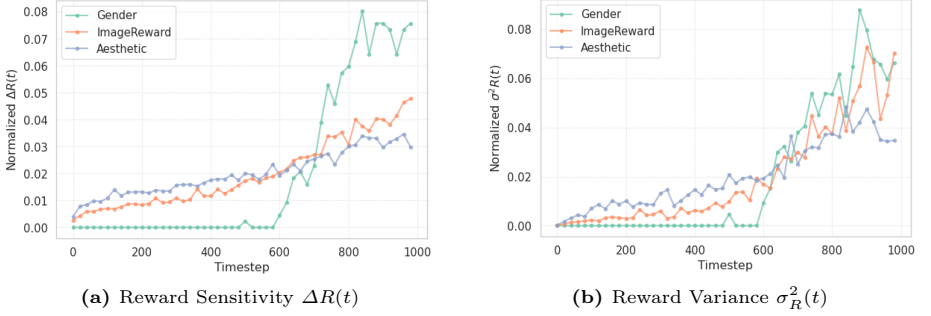


Fig. 2: Reward behavior across diffusion timesteps. (a) Sensitivity $\Delta R(t)$ measures how strongly each reward reacts to corrupting an image up to timestep t . (b) Variance $\sigma^2 R(t)$ captures the stability of reward evaluations under repeated corruptions. Curves are reported for three reward functions—gender score (*green*), ImageReward (*orange*), and LAION Aesthetic Predictor (*blue*)—evaluated on reconstructions generated by a DDPM trained on CelebA-HQ [20].

number of timesteps $T = 50$, following the DDIM scheduler approach [38], in which fewer timesteps are sampled from the original noise schedule (i.e., a total of 1000 steps in our particular model). For that, we employ three representative reward functions: (a) a gender classification score [10], (b) ImageReward [44], which measures text-to-image human preference, and (c) the LAION Aesthetic Predictor [36], which quantifies the aesthetic appeal of the images. Fig. 2 reports reward sensitivity $\Delta R(t)$ and reward variance $\sigma^2 R(t)$ across diffusion timesteps for these three functions.

The three rewards exhibit distinct timestep-dependent behaviors. For the gender classification reward, both metrics remain near zero over most of the trajectory: corrupting an image up to relatively high noise levels ($t \approx 600$) does not alter the gender of the reconstructions. Only at late timesteps—when the image is heavily degraded—do sensitivity and variance increase sharply, indicating that gender-relevant cues are encoded primarily in coarse, low-frequency structure.

In contrast, the aesthetic and ImageReward signals exhibit broader dependence on the diffusion trajectory. For the aesthetic predictor, both sensitivity and variance increase approximately linearly with the timestep, indicating that aesthetic-relevant information is progressively lost as corruption increases. Notably, even early, low-noise corruption leads to measurable changes in reconstructed aesthetic scores, suggesting that aesthetic cues are distributed across both early and late denoising stages. ImageReward follows a similar trend at early timesteps but transitions to a steeper increase—particularly in variance—beyond a corruption threshold. Given that the ImageReward prompt emphasizes overall visual plausibility rather than fine-grained detail, this behavior is consistent with a sharper degradation once coarse, semantically meaningful structure begins to collapse.

Taken together, these results highlight that each reward function depends on a different region of the diffusion trajectory. This heterogeneity underscores

the importance of accounting for reward–timestep interactions when designing reward-based fine-tuning and sampling strategies.

5.2 Policy-Gradient Fine-Tuning Setup

We initialize the policy with a publicly available DDPM [18] trained on CelebA-HQ [20]. Fine-tuning aims to increase the generation of male-presenting faces while preserving alignment with human preference. For a generated image x_0 , the reward is defined as

$$r(x_0) = \text{IR}(x_0) + 2 \cdot \text{Gender}(x_0), \quad (8)$$

where IR denotes ImageReward [44] evaluated with a fixed text prompt, and $\text{Gender}(x_0)$ is defined as a binary score obtained by thresholding the predicted male probability at 0.8 from a pretrained classifier [10].

All experiments follow the policy-gradient formulation in Sec. 3, varying only the timestep sampling strategy. Each configuration is trained for 100 iterations and repeated twice under an identical training setup, as the goal is to compare relative efficiency rather than maximize absolute performance. Full optimization details are provided in the Supplementary Material.

For sampling, we use a DDIM scheduler [38] with $T = 50$ denoising steps, sampled from the original noise schedule (i.e., 1000 steps). Our Full-Trajectory baseline follows the standard policy-gradient update over the entire denoising trajectory (Eq. 2). In contrast, all our sampling-based strategies evaluate 20 timesteps per update, drawn independently from the corresponding timestep distribution. This fixed-timestep budget ensures a controlled comparison under reduced computation. Fig. 3 summarizes the sampling distributions used across all strategies, which are described in detail in the following sections.

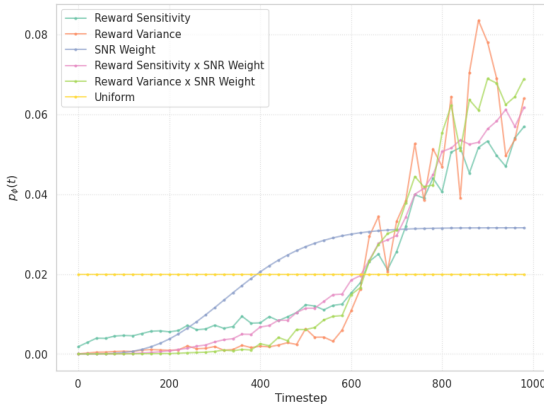


Fig. 3: Timestep sampling distributions evaluated for fine-tuning the CelebA-HQ DDPM model under the reward defined in Eq. 8.

5.3 Sampling Strategies

We present a systematic progression of sampling strategies, starting from reward-agnostic baselines—full-trajectory, uniform sampling, and SNR-based weighting—and culminating in our reward-aware and hybrid sampling formulations.

Full-Trajectory. This strategy uses the standard policy-gradient update described in Sec. 5.2, aggregating gradients across all $T = 50$ timesteps.

Uniform sampling. We sample timesteps uniformly, reducing the number of evaluated steps from 50 to 20 per update while treating all steps identically. Any deviation from the full estimator, therefore, reflects the effect of subsampling alone.

SNR-based sampling. This strategy (Sec. 4.2) instead weights timesteps according to Eq. 5, using the recommended hyperparameters $k = 1$ and $\gamma = 1$ from [5]. Although still reward-agnostic, this strategy introduces a structural bias based on signal-to-noise characteristics of the diffusion process, reflecting the intuition that timesteps contribute unequally to optimization.

Reward-Aware Sampling. We construct sampling distributions from the Reward Sensitivity and Reward Variance metrics defined in Sec. 4.3. They are estimated using $N = 10$ samples with $M = 10$ reconstructions per timestep. These estimates yield timestep-indexed weighting functions, denoted $w_{\Delta R}(t)$ and $w_{\sigma^2 R}(t)$, which are normalized as described in Sec. 4.1.

Hybrid SNR and Reward-Aware Sampling. We evaluate hybrid strategies that multiplicatively combine SNR with our reward-aware signals, Reward Sensitivity ($w_{\Delta R}(t)$), and Reward Variance ($w_{\sigma^2 R}(t)$):

$$w_{\text{SNR} \times \Delta R}(t) = w_{\text{SNR}}(t) w_{\Delta R}(t), \quad w_{\text{SNR} \times \sigma^2 R}(t) = w_{\text{SNR}}(t) w_{\sigma^2 R}(t).$$

5.4 Comparative Analysis of Sampling Strategies

Quantitative Results. As illustrated in Fig. 4, Uniform sampling (blue curve) effectively recovers the Full-Trajectory baseline (orange) performance despite a 60% reduction in timestep evaluation, incurring $\approx 23\%$ reward degradation. This suggests that large segments of the denoising trajectory are redundant for optimization, allowing for a favorable compute–performance trade-off.

Constrained to the same 20-step budget, SNR weighting (yellow) yields accelerated convergence and superior final rewards compared to Uniform sampling (blue). Notably, it also substantially surpasses the Full-Trajectory baseline (orange), achieving both higher reward and faster convergence. This early divergence in learning curves confirms that structurally informative timesteps possess

disproportionate optimization value, even when identified via reward-agnostic structural profiles.

While Sensitivity sampling (magenta) remains competitive with SNR (yellow), Variance-based (green) prioritization consistently achieves faster convergence and higher asymptotic rewards. This indicates that reward variability is a more robust signal of timestep importance than sensitivity alone. Crucially, both reward-aware metrics significantly outperform Full-Trajectory (orange) and Uniform (blue) baselines, validating the necessity of principled timestep allocation.

Finally, our hybrid strategies demonstrate synergy, outperforming their isolated counterparts. The Sensitivity $\times$ SNR combination (beige curve) improves over isolated Sensitivity-based sampling (magenta), achieving both the highest final reward and the largest speedup. Similarly, the Variance $\times$ SNR hybrid (gray) surpasses Variance-based weighting (green), matching the best observed convergence speed while maintaining comparable final performance.

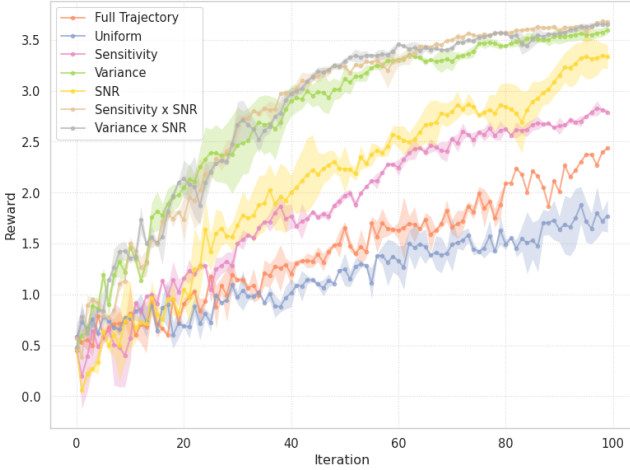


Fig. 4: Comparison of all timestep sampling strategies. Each curve shows the evolution of the total reward throughout fine-tuning under a different sampling distribution, enabling a direct comparison between: i) reward-agnostic baselines: Full-Trajectory (orange), Uniform (blue), SNR (yellow); ii) our reward-aware strategies: Sensitivity (magenta), Variance (green); and iii) our hybrid SNR–reward combinations: Sensitivity $\times$ SNR (beige), Variance $\times$ SNR (gray).

Table 1 provide a consolidated comparison across all sampling strategies. The results exhibit a consistent hierarchy: uniform subsampling largely preserves the behavior of the full estimator under reduced computation; SNR-based weighting substantially accelerates convergence within the same reduced timestep budget; reward-aware methods—particularly variance-based weighting—yield additional efficiency gains; and the hybrid schemes deliver the largest measured speedups, reaching over $3\times$ acceleration relative to the full-trajectory estimator.

Qualitative Results. Fig. 5 presents representative outputs illustrating sampling strategies. Models that account for the non-uniform contribution of time-steps generate images more aligned with the objective (i.e., generation of males aligned with human preference) under the same training budget. Notably, the strategies that achieve the highest reward (i.e., Variance, Sensitivity $\times$ SNR, Variance $\times$ SNR) exhibit reduced visual diversity, producing similar samples, consistent with prior observations from the literature [6,49]. Investigating mechanisms that better balance reward maximization and output diversity remains an important direction for future work.

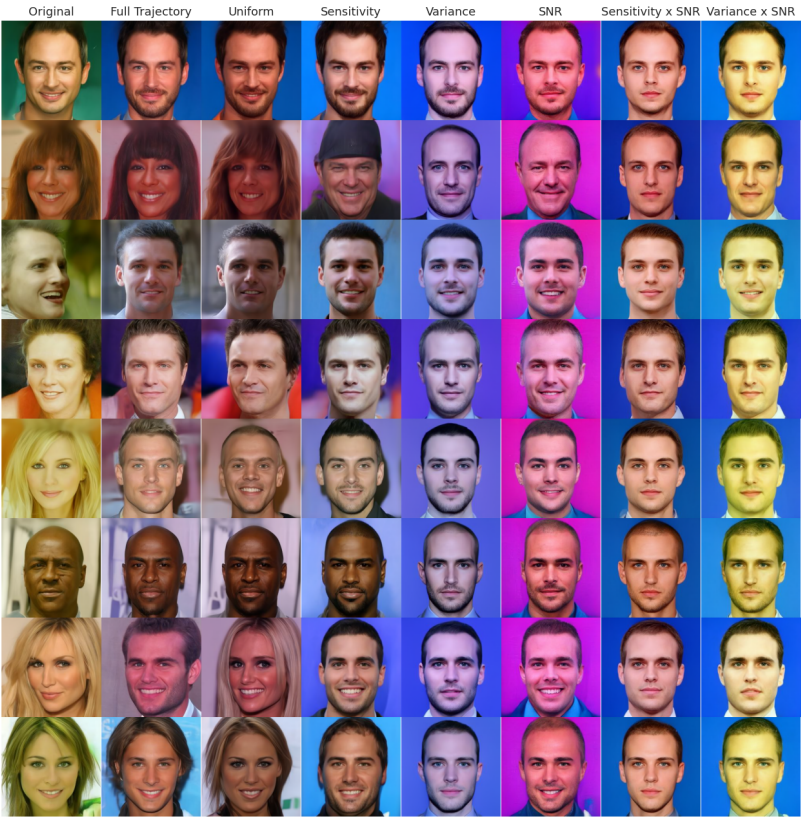


Fig. 5: Qualitative comparison of generated outputs across sampling strategies. Each column displays samples produced by a model fine-tuned with a different timestep sampling scheme, with the pretrained model shown in the first column. All images are generated using the same random seed, isolating the effect of the fine-tuning strategy on the final outputs. The figure highlights systematic visual shifts introduced by the various weighting methods.

Table 1: Performance summary for all timestep sampling strategies. For each method, we report the final reward, the maximum reward achieved during training, and the speedup relative to the Full-Trajectory timestep estimator. Speedup is computed as the ratio between the iteration at which the Full-Trajectory timestep estimator reaches its best reward and the iteration at which the evaluated method reaches the same reward level.

Method	Final Reward	Best Reward	Speedup
Full-Trajectory	2.43	2.43	1.0×
Uniform	1.77	1.88	—
Sensitivity	2.79	2.83	1.50×
Variance	3.59	3.59	3.30×
SNR	3.34	3.35	1.70×
Sensitivity × SNR	3.68	3.68	3.41×
Variance × SNR	3.65	3.66	3.41×

6 Conclusion

Policy-gradient fine-tuning enables diffusion models to optimize arbitrary, non-differentiable reward functions, but its standard formulation aggregates gradients from all denoising timesteps at every update, leading to substantial computational and convergence inefficiencies. In this work, we analyze how individual timesteps contribute to the policy-gradient objective and show that this full-trajectory evaluation is neither necessary nor optimal.

We introduced two novel reward-dependent importance measures — Reward Sensitivity and Reward Variance — that quantify the empirical influence of each timestep on downstream rewards. These metrics provide a principled way to characterize timestep relevance under a given objective and can be directly incorporated into timestep sampling. Empirically, variance-based sampling yields substantial gains in convergence speed and final reward, while sensitivity-based sampling remains competitive with structural SNR-based weighting.

In a systematic comparison of sampling strategies, we show, within a controlled experimental setting, that informed timestep allocation reduces cost and accelerates convergence relative to the full-trajectory estimator, while increasing rewards. SNR-based weighting already provides meaningful improvements, reward-variance sampling further enhances efficiency, and combining SNR and reward-aware signals yields the largest speedups in our setting.

Overall, our results demonstrate that timestep contributions in policy-gradient fine-tuning are both non-uniform and reward-dependent, and that leveraging this structure offers a principled and practical path toward more efficient reinforcement learning for diffusion models. While our empirical evaluation is restricted to a particular task, to our knowledge, this is the first work to tackle this specific aspect of diffusion policy-gradient fine-tuning — timestep- and reward-aware optimization — through a principled approach. Therefore, it leaves broader evaluation across additional benchmarks and reward functions as an important direction for future work.

References

1. Black, K., Janner, M., Du, Y., Kostrikov, I., Levine, S.: Training diffusion models with reinforcement learning. arXiv preprint arXiv:2305.13301 (2023)
2. Chefer, H., Alaluf, Y., Vinker, Y., Wolf, L., Cohen-Or, D.: Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models. ACM transactions on Graphics (TOG) **42**(4), 1–10 (2023)
3. Chen, C., Wang, A., Wu, H., Liao, L., Sun, W., Yan, Q., Lin, W.: Enhancing diffusion models with text-encoder reinforcement learning. In: European Conference on Computer Vision. pp. 182–198. Springer (2024)
4. Chen, M., Laina, I., Vedaldi, A.: Training-free layout control with cross-attention guidance. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision. pp. 5343–5353 (2024)
5. Choi, J., Lee, J., Shin, C., Kim, S., Kim, H., Yoon, S.: Perception prioritized training of diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 11472–11481 (2022)
6. Clark, K., Vicol, P., Swersky, K., Fleet, D.J.: Directly fine-tuning diffusion models on differentiable rewards. arXiv preprint arXiv:2309.17400 (2023)
7. Dai, X., Hou, J., Ma, C.Y., Tsai, S., Wang, J., Wang, R., Zhang, P., Vandenhende, S., Wang, X., Dubey, A., et al.: Emu: Enhancing image generation models using photogenic needles in a haystack. arXiv preprint arXiv:2309.15807 (2023)
8. Dhariwal, P., Nichol, A.: Diffusion models beat gans on image synthesis. Advances in neural information processing systems **34**, 8780–8794 (2021)
9. Dong, H., Xiong, W., Goyal, D., Zhang, Y., Chow, W., Pan, R., Diao, S., Zhang, J., Shum, K., Zhang, T.: Raft: Reward ranked finetuning for generative foundation model alignment. arXiv preprint arXiv:2304.06767 (2023)
10. Dwikifirdaus, R.: Gender Classification. <https://huggingface.co/rizvandwiki/gender-classification> (2022)
11. Everaert, M.N., Bocchio, M., Arpa, S., Süssstrunk, S., Achanta, R.: Diffusion in style. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 2251–2261 (October 2023)
12. Fan, Y., Watkins, O., Du, Y., Liu, H., Ryu, M., Boutilier, C., Abbeel, P., Ghavamzadeh, M., Lee, K., Lee, K.: Dpok: Reinforcement learning for fine-tuning text-to-image diffusion models. Advances in Neural Information Processing Systems **36**, 79858–79885 (2023)
13. Feng, W., He, X., Fu, T.J., Jampani, V., Akula, A., Narayana, P., Basu, S., Wang, X.E., Wang, W.Y.: Training-free structured diffusion guidance for compositional text-to-image synthesis. arXiv preprint arXiv:2212.05032 (2022)
14. Feng, W., Zhu, W., Fu, T.j., Jampani, V., Akula, A., He, X., Basu, S., Wang, X.E., Wang, W.Y.: Layoutgpt: Compositional visual planning and generation with large language models. Advances in Neural Information Processing Systems **36**, 18225–18250 (2023)
15. Guo, Z., Liu, J., Wang, Y., Chen, M., Wang, D., Xu, D., Cheng, J.: Diffusion models in bioinformatics and computational biology. Nature reviews bioengineering **2**(2), 136–154 (2024)
16. Gupta, S., Ahuja, C., Lin, T.Y., Roy, S.D., Oosterhuis, H., de Rijke, M., Shukla, S.N.: A simple and effective reinforcement learning method for text-to-image diffusion fine-tuning. arXiv preprint arXiv:2503.00897 (2025)
17. Hang, T., Gu, S., Li, C., Bao, J., Chen, D., Hu, H., Geng, X., Guo, B.: Efficient diffusion training via min-snr weighting strategy. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 7441–7451 (2023)

18. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. *Advances in neural information processing systems* **33**, 6840–6851 (2020)
19. Ho, J., Salimans, T., Gritsenko, A., Chan, W., Norouzi, M., Fleet, D.J.: Video diffusion models. *Advances in neural information processing systems* **35**, 8633–8646 (2022)
20. Karras, T., Aila, T., Laine, S., Lehtinen, J.: Progressive growing of gans for improved quality, stability, and variation. *arXiv preprint arXiv:1710.10196* (2017)
21. Karras, T., Aittala, M., Aila, T., Laine, S.: Elucidating the design space of diffusion-based generative models. *Advances in neural information processing systems* **35**, 26565–26577 (2022)
22. Kim, M., Ki, D., Shim, S.W., Lee, B.J.: Adaptive non-uniform timestep sampling for diffusion model training. *arXiv preprint arXiv:2411.09998* (2024)
23. Kingma, D., Salimans, T., Poole, B., Ho, J.: Variational diffusion models. *Advances in neural information processing systems* **34**, 21696–21707 (2021)
24. Lee, K., Liu, H., Ryu, M., Watkins, O., Du, Y., Boutilier, C., Abbeel, P., Ghavamzadeh, M., Gu, S.S.: Aligning text-to-image models using human feedback. *arXiv preprint arXiv:2302.12192* (2023)
25. Li, S., Kallidromitis, K., Gokul, A., Kato, Y., Kozuka, K.: Aligning diffusion models by optimizing human utility. In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems* (2024)
26. Li, W., Su, X., Han, Y., You, S., Huang, T., Xu, C.: Adaptive training meets progressive scaling: Elevating efficiency in diffusion models. *arXiv e-prints pp. arXiv–2312* (2023)
27. Li, X., Thickstun, J., Gulrajani, I., Liang, P.S., Hashimoto, T.B.: Diffusion-lm improves controllable text generation. *Advances in neural information processing systems* **35**, 4328–4343 (2022)
28. Li, Y., Liu, H., Wu, Q., Mu, F., Yang, J., Gao, J., Li, C., Lee, Y.J.: Gligen: Open-set grounded text-to-image generation. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 22511–22521 (2023)
29. Lian, L., Li, B., Yala, A., Darrell, T.: Llm-grounded diffusion: Enhancing prompt understanding of text-to-image diffusion models with large language models. *arXiv preprint arXiv:2305.13655* (2023)
30. Luo, S., Hu, W.: Diffusion probabilistic models for 3d point cloud generation. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 2837–2845 (2021)
31. Ma, Q., Ning, X., Liu, D., Niu, L., Zhang, L.: Decouple-then-merge: Finetune diffusion models as multi-task learning. In: *Proceedings of the Computer Vision and Pattern Recognition Conference*. pp. 23281–23291 (2025)
32. Miao, Z., Wang, J., Wang, Z., Yang, Z., Wang, L., Qiu, Q., Liu, Z.: Training diffusion models towards diverse image generation with reinforcement learning. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 10844–10853 (2024)
33. Moon, T., Choi, M., Lee, G., Ha, J.W., Lee, J.: Fine-tuning diffusion models with limited data. In: *NeurIPS 2022 Workshop on Score-Based Methods* (2022), <https://openreview.net/forum?id=0J6afk9DqrR>
34. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 10684–10695 (2022)
35. Ruiz, N., Li, Y., Jampani, V., Pritch, Y., Rubinstein, M., Aberman, K.: Dream-booth: Fine tuning text-to-image diffusion models for subject-driven generation. In:

- Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 22500–22510 (2023)
36. Schuhmann, C., Beaumont, R., Vencu, R., Gordon, C., Wightman, R., Cherti, M., Coombes, T., Katta, A., Mullis, C., Wortsman, M., et al.: Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in neural information processing systems* **35**, 25278–25294 (2022)
37. Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., Ganguli, S.: Deep unsupervised learning using nonequilibrium thermodynamics. In: *International conference on machine learning*. pp. 2256–2265. *pmlr* (2015)
38. Song, J., Meng, C., Ermon, S.: Denoising diffusion implicit models. In: *International Conference on Learning Representations* (2020)
39. Sutton, R., Barto, A.: *Reinforcement Learning, second edition: An Introduction*. Adaptive Computation and Machine Learning series, MIT Press (2018), <https://books.google.com.br/books?id=sWVODwAAQBAJ>
40. Wallace, B., Dang, M., Rafailov, R., Zhou, L., Lou, A., Purushwalkam, S., Ermon, S., Xiong, C., Joty, S., Naik, N.: Diffusion model alignment using direct preference optimization. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 8228–8238 (2024)
41. Wang, K., Shi, M., Zhou, Y., Li, Z., Yuan, Z., Shang, Y., Peng, X., Zhang, H., You, Y.: A closer look at time steps is worthy of triple speed-up for diffusion model training. In: *Proceedings of the Computer Vision and Pattern Recognition Conference*. pp. 12934–12944 (2025)
42. Williams, R.J.: Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine learning* **8**(3), 229–256 (1992)
43. Wu, X., Sun, K., Zhu, F., Zhao, R., Li, H.: Human preference score: Better aligning text-to-image models with human preference. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 2096–2105 (2023)
44. Xu, J., Liu, X., Wu, Y., Tong, Y., Li, Q., Ding, M., Tang, J., Dong, Y.: Imagereward: Learning and evaluating human preferences for text-to-image generation. *Advances in Neural Information Processing Systems* **36**, 15903–15935 (2023)
45. Xu, T., Mi, P., Wang, R., Chen, Y.: Towards faster training of diffusion models: An inspiration of a consistency phenomenon. *arXiv preprint arXiv:2404.07946* (2024)
46. Yang, K., Tao, J., Lyu, J., Ge, C., Chen, J., Li, Q., Shen, W., Zhu, X., Li, X.: Using human feedback to fine-tune diffusion models without any reward model. URL: <http://arxiv.org/abs/2311.13231> (2024)
47. Yang, S., Chen, Y., Wang, L., Liu, S., Chen, Y.C.: Denoising diffusion step-aware models. In: *The Twelfth International Conference on Learning Representations* (2023)
48. Yuan, H., Huang, K., Ni, C., Chen, M., Wang, M.: Reward-directed conditional diffusion: Provable distribution estimation and reward improvement. *Advances in Neural Information Processing Systems* **36**, 60599–60635 (2023)
49. Zhang, Y., Tzeng, E., Du, Y., Kislyuk, D.: Large-scale reinforcement learning for diffusion models. In: *European Conference on Computer Vision*. pp. 1–17. Springer (2024)
50. Zhao, S., Gupta, D., Zheng, Q., Grover, A.: d1: Scaling reasoning in diffusion large language models via reinforcement learning. *arXiv preprint arXiv:2504.12216* (2025)
51. Zheng, T., Geng, C., Jiang, P.T., Wan, B., Zhang, H., Chen, J., Wang, J., Li, B.: Non-uniform timestep sampling: Towards faster diffusion model training. In: *Proceedings of the 32nd ACM International Conference on Multimedia*. pp. 7036–7045 (2024)

52. Zheng, T., Jiang, P.T., Wan, B., Zhang, H., Chen, J., Wang, J., Li, B.: Beta-
tuned timestep diffusion model. In: European Conference on Computer Vision. pp.
114–130. Springer (2024)